Test 2 – Grade 8

Name: _____
Grade / Class: 8

Section A: English Language & Literature

1. Reading Comprehension
Read the passage and answer the questions:

"Alex decided to volunteer at a local animal shelter during the weekend. He helped feed the animals, clean their cages, and learn about the responsibilities of pet care. Through this experience, Alex realized the importance of compassion and teamwork."

a) What activity did Alex participate in during the weekend?
b) List three tasks Alex performed at the shelter.
c) What lesson did Alex learn from this experience?
d) Write the passage in your own words in three sentences.

2. Grammar & Vocabulary
a) Change the following sentence into negative form:
"The students finished their assignments on time."

b) Identify the adverbs in the sentence:
"The teacher spoke clearly and patiently during the lecture."

c) Fill in the blank with the correct word:
"The weather forecast says it will be _____ tomorrow. (rainy / raining)"

d) Rewrite this sentence in direct speech:
"He asked if the library was open today."

3. Writing
Write a formal letter of 120–150 words to the principal requesting permission to organize a school science fair.

Section B: Mathematics

1. Algebra
a) Solve for y: 7y+3=457y + 3 = 457y+3=45
b) Expand: (x+5)(x−3)(x + 5)(x - 3)(x+5)(x−3)
c) Simplify: 4m2n−6mn2+3m2n4m^2n - 6mn^2 + 3m^2n4m2n−6mn2+3m2n

2. Geometry
a) Find the perimeter of a rectangle with length 18 cm and width 12 cm.
b) A cone has radius 7 cm and height 24 cm. Find its volume.

3. Ratio, Proportion & Percentage
a) A jacket costs $150. It is sold with a 20% discount. Find the discounted price.
b) If 5 notebooks cost $25, what is the cost of 15 notebooks?

4. Data Handling
The scores of 12 students in a test are: 14, 18, 15, 20, 16, 17, 18, 15, 19, 16, 17, 14.
a) Calculate mean, median, and mode.
b) Draw a simple bar graph representing these scores.

Section C: Science

1. Biology
a) What is the main function of the human liver?
b) Explain the process of respiration in plants.
c) State two differences between amphibians and reptiles.

2. Physics
a) A ball falls from a height of 45 m. Find the time it takes to reach the ground (ignore air resistance, g = 9.8 m/s²).
b) Define work and give an example.
c) What is the difference between reflection and refraction of light?

3. Chemistry
a) Identify whether the following are elements, compounds, or mixtures:

- **Carbon dioxide**
- **Iron**
- **Sand and sugar mixture**

b) Write the balanced chemical equation for the reaction between zinc and hydrochloric acid.

4. Environmental Science
a) Name two major water pollutants and their sources.
b) Suggest two ways to reduce plastic waste.

Section D: Social Studies

1. History
a) Name two inventions of the 19th century that changed transportation.
b) How did the steam engine affect industrial growth?
c) Name the countries forming the Axis powers in World War II.

2. Geography
a) Define longitude and explain its importance.
b) Name the highest mountain in Africa and the continent it is located in.
c) Explain two effects of desertification.

3. Civics / Political Science
a) Compare a republic and a dictatorship.
b) List two duties of citizens in a democratic society.

4. Economics
a) Define inflation and deflation.
b) What is the difference between goods and services?

Section E: ICT / Computer Studies

1. Computer Basics
a) What is an operating system? Give one example.
b) Name one storage device and one input device.

2. Applications
a) Which spreadsheet formula calculates the average of numbers?
b) Explain one use of word processing software in school.

3. Programming Basics
a) Define variable and give an example in Python.
b) Write a small Python code to display the sum of two numbers.

4. Cyber Awareness
a) Explain phishing and one way to avoid it.
b) Give two guidelines for responsible social media use.

Section F: Islamiyat

1. Quran & Hadith
a) Name two virtues emphasized in the Quran.
b) Narrate one Hadith that highlights honesty in daily life.

2. Aqidah / Belief
a) List the six articles of Iman (faith) in Islam.
b) Explain the concept of Risalah (prophethood).

3. Fiqh / Worship
a) What is the significance of Zakat in Islam?
b) Mention the five pillars of Islam and briefly explain one.

4. Islamic Ethics & Morality
a) Explain the importance of justice in Islam.
b) How does Islam guide humans to respect animals and the environment?

Section G: Arts & Music

1. Visual Arts
a) Explain symmetry in drawing.
b) Name two secondary colors.

2. Music
a) Define tempo and pitch.
b) Give an example of a percussion instrument from any culture.

Section H: Health & Physical Education

1. Physical Fitness
a) Name two exercises to increase endurance.
b) List one Olympic sport and one recreational activity.

2. Health & Hygiene
a) Why is regular exercise important for health?
b) Mention two ways to prevent infectious diseases.

3. Safety & Environment
a) Name one safety precaution during a fire drill.
b) Suggest one method to save electricity at home.